

Sorting:-

Sorting is the process of arranging data in a specific order, such as ascending or descending order, based on one or more values of attributes.

Ex:- Arranging students according to their marks from highest to lowest.

Grouping:-

Grouping is the process of organizing data into categories or groups based on a common attribute.

It helps in analyzing data within specific categories.

Ex:- Grouping students according to their departments such as CSE, ECE, and IT.

Ranking:-

Ranking is the process of organizing data into categories or groups based on their values of performance.

Ex:- Assign Rank 1 to the student with the highest marks.

Plotting:-

Plotting is the graphical representation of data using charts, graphs, or visualizations. It helps in understanding patterns, trends, and relationships in data.

Ex:- Representing student marks using a bar chart or line graph.

1. Sorting

```
print " / n sorted by marks :")
```

```
sorted_df = df.sort_values(by='marks', ascending = false)
```

```
print (sorted_df)
```

2. Grouping

print ("Average Marks by department:")

grouped - df = df.groupby("Department")["Marks"].mean()

print (grouped - df)

3. Ranking

df["Rank"] = df["Marks"].rank(ascending=False, method="dense")

print ("Students with Rank:")

print (df.sort_values(by="Rank"))

4. Plotting

Bar chart of students marks.

plt.figure(figsize=(8,5))

plt.bar(df["Name"], df["Marks"])

plt.title("Student Marks")

plt.xlabel("Student name")

plt.ylabel("Marks")

plt.tight_layout()

Plot Average marks by department

grouped - df plot (kind="bar")

plt.title("Average Marks by Department")

plt.xlabel("Average Marks")

plt.ylabel("Department")

plt.show()

Output ==

Name Dept Marks

5. Pathina ECE 95

2. Chitra CSE 92

3. Divya ECE 88

0. Arun CSE 85

1. Bala ECE 78

4. Ezhil CSE 75

Average

CSE = 84

ECE = 87

Students with rank.

5. Pathina ECE 95 1

2. Chitra CSE 92 2

3. Divya ECE 88 3

0. Arun CSE 85 4

4. Bala ECE 78 5

